

BHS Symposium 2026

Minds in Motion

Program schedule

Learning Studio, Experimental Medicine Building, NTU, Singapore

1-2 July 2026 | Wednesday-Thursday

DAY 1

Wed, 1 Jul 2026

Day 1

● Talks ● Workshops ● Breaks ● Visits

MORNING / PART 1

0800	MEAL Breakfast & Registration Arrive, collect materials, and meet fellow participants.
0900	OPENING Prof Annabel Chen Nanyang Technological University, Singapore Opening Address Welcome and symposium kickoff.
0910	TALK Dr Hsu Tzu-Yu National Central University, Taiwan Neural correlates of switching between internally and externally oriented thoughts Internal/external cognition, EEG, fMRI, rumination, and attentional switching.
0955	TALK Dr Byung-Doh Oh Nanyang Technological University, Singapore To model human linguistic prediction, make LLMs less superhuman Human-like memory constraints and reading behavior in language models.
1040	BREAK Short Break 15 minutes

AFTERNOON / PART 2

1055	PITCH + Q&A Students' Pitch on BrainHack Projects + Q&A Project pitches and discussion.
1215	MEAL Lunch Break 1 hour 15 minutes
1330	PITCH + Q&A Students' Pitch on BrainHack Projects + Q&A Afternoon project pitches and discussion.
1500-1530	BREAK Tea Break & Networking Refreshments and informal networking.
1530	VISIT CoNIC visit and Campus tour Guided visit and campus tour.

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DAY 2

Thu, 2 Jul 2026

Day 2

● Talks ● Workshops ● Breaks ● Visits

MORNING / PART 1

0900	TALK Dr Stefano V. Albrecht Nanyang Technological University, Singapore Multi-agent Reinforcement Learning MARL foundations, algorithmic ideas, and applications in cognitive science.
0945	TALK Dr Charly Hugo Alexandre Billaud Nanyang Technological University, Singapore Cortical and subcortical surfaces shaping human behaviour Brain shape, ageing, cognition, behaviour, and open neuroimaging tools.
1030	BREAK Morning Tea Break 15 minutes
1045-1130	TALK Dr Junhong Yu Nanyang Technological University, Singapore The Resting Mind: How Spontaneous Thought Shapes Brain Connectivity, Brain States, and Behavior Resting-state thought, connectivity, brain states, and behavior.
1130	PRESENTATION Amanda Lin National Taiwan University; Ting-Hsin Yen National Central University; Tiffany Hsin-Yu Lin NTU Presentation by BrainHack School Teaching Assistants

AFTERNOON / PART 2

1200	MEAL Lunch Break 1 hour
1300	WORKSHOP Dr Kevin Chun-Hsien Hsu Institute of Cognitive Neuroscience, National Central University MEG Workshop MEG signals, cognitive/language applications, and data workflow demonstration.
1400	BREAK Break 15 minutes
1415	WORKSHOP Dr Niall Duncan's Team Taipei Medical University, Taiwan Open Science Workshop Dr Elizaveta Baranova: Unintentional scientific misconduct; Trinh Thi Thuy Ngo: Introduction to multiverse analysis.
1545-1600	CLOSING Dr Joshua Goh National Taiwan University, Taiwan Closing Address & Prize Giving Ceremony Closing remarks and prize presentation.

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Talk titles, speakers, and abstracts

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Day 1 - Talk Titles, Speakers, and Abstracts

DAY 1 TALK

Dr Hsu Tzu-Yu

National Central University, Taiwan



Title: Neural correlates of switching between internally and externally oriented thoughts

The capacity to flexibly shift between processing external sensory information and engaging in internal thought is essential for everyday functioning. In major depressive disorder (MDD), this balance is disrupted - patients become trapped in repetitive, self-focused rumination at the expense of engaging with their environment. We present converging EEG and fMRI evidence demonstrating that internal and external cognitive states are supported by dissociable neural signatures, and critically, that switching between them is not symmetric. Resting-state neural dynamics linked to rumination emerge specifically during internally oriented states, while transitioning toward external processing recruits additional occipital and subcortical regions that are not equivalently engaged when shifting inward. These findings suggest that the "stickiness" of internal rumination in depression may partly reflect a failure of the neural gating mechanisms required to redirect attention outward. Understanding this switching architecture opens potential avenues for targeted neuromodulation strategies aimed at restoring flexible transitions between internal and external cognition in affective disorders.

DAY 1 TALK

Dr Byung-Doh Oh

Nanyang Technological University, Singapore



Title: To model human linguistic prediction, make LLMs less superhuman

When we read, we make predictions about upcoming words; these predictions influence our reading behavior. The success of large language models (LLMs), which, like humans, make predictions about upcoming words, has motivated their use as models of human linguistic prediction. Surprisingly, in the last few years, as LLMs' ability to predict the next word has improved, their ability to explain reading behavior has declined. In this talk, I argue that this is because current LLMs can predict upcoming words much better than human readers; this 'superhumanness' is driven by LLMs' extensive training data, stronger long-term memory of training examples, and stronger short-term memory. To further our understanding of human language processing, I advocate for creating LLMs with human-like memory, and outline directions for achieving this goal.

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Day 2 - Talk Titles, Speakers, and Abstracts

DAY 2 TALK

Dr Stefano V. Albrecht

Nanyang Technological University, Singapore

Title: Multi-agent Reinforcement Learning

Multi-agent reinforcement learning (MARL) enables systems of autonomous agents to learn how to interact and collaborate. MARL is a highly active area of research at the intersection of AI, game theory, and deep learning, with a range of applications in human-AI interaction, multi-robot systems, and social/economic sciences. For cognitive science and brain modelling, MARL methodology could potentially provide new insights and modelling approaches for system-level and neural-level interaction in brains. This lecture will provide a short introduction to the main models and algorithmic ideas at the foundation of MARL. The lecture will be based on the MIT Press textbook "Multi-Agent Reinforcement Learning: Foundations and Modern Approaches", which is freely available at www.marl-book.com.



DAY 2 TALK

Dr Charly Hugo Alexandre Billaud

Nanyang Technological University, Singapore

Title: Cortical and subcortical surfaces shaping human behaviour

With the popularization of advanced neuroimaging analyses methods, the brain and its structures are frequently represented as 3D mesh surfaces. These representations inform us on how cortical and subcortical regions are shaped, gyrified, curved, elongated, thickening and more. These focal, subtle geometric features evolve with ageing and influence cognition and behaviour. Freely available, user-friendly tools to produce them, and to statistically analyse the relation between brain shape and behaviours will be presented with practical demonstrations on open data from ageing populations.



DAY 2 TALK

Dr Junhong Yu

Nanyang Technological University, Singapore

Title: The Resting Mind: How Spontaneous Thought Shapes Brain Connectivity, Brain States, and Behavior

What do people think about during resting-state fMRI scans, and how do these thoughts shape brain activity and psychological traits? Although resting-state fMRI is widely used to study brain-behavior relationships, the subjective experiences occurring during rest are often overlooked. Using resting-state scans annotated with thought-content ratings, I show that various dimensions of spontaneous cognition are associated with both static functional connectivity patterns and dynamic coactivation-defined brain states. Furthermore, these thought dimensions significantly mediated relationships between resting-state connectivity and a broad range of behavioral phenotypes. Together, these findings suggest that spontaneous cognition is not merely noise during rest, but meaningfully shapes the spatiotemporal organization of the resting-state brain and explains observed brain-behavior relationships.



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Day 2 - Workshop Titles, Speakers, and Abstracts

DAY 2 WORKSHOP

Dr Kevin Chun-Hsien Hsu

Institute of Cognitive Neuroscience, National Central University

Title: MEG Workshop

This talk will introduce magnetoencephalography (MEG), covering the basic physics and physiology underlying MEG signals and their application in studying cognition and language processes. In addition, a brief demonstration of the MEG data workflow, from acquisition to analysis.



OPEN SCIENCE WORKSHOP

Trinh Thi Thuy Ngo

Taipei Medical University, Taiwan

Title: An introduction to multiverse analysis

When analysing data we have a huge array of choices to be made throughout the process. The particular path that we take through these choices influences the outcome that we get. These high degrees of researcher degrees of freedom have the potential to lead to unreliable conclusions being made and reported. I will introduce the idea of a multiverse analysis and explain how this may be one approach to dealing with this reliability issue. I will demonstrate a simple implementation of such an analysis to give participants a foundation for potentially employing such an analysis themselves.



OPEN SCIENCE WORKSHOP

Dr Elizaveta Baranova

Taipei Medical University, Taiwan

Title: Unintentional scientific misconduct: Ways of creating errors and of avoiding them

As how we do science progresses the range of useful tools available to us always increases. At the same time, the opportunities for us to make a mistake also seems to increase. In this informal workshop I will talk about different potential sources of errors and some principles that can help us avoid them (mostly). This will include our approach to data, code, and new problems that may arise from AI tools.

